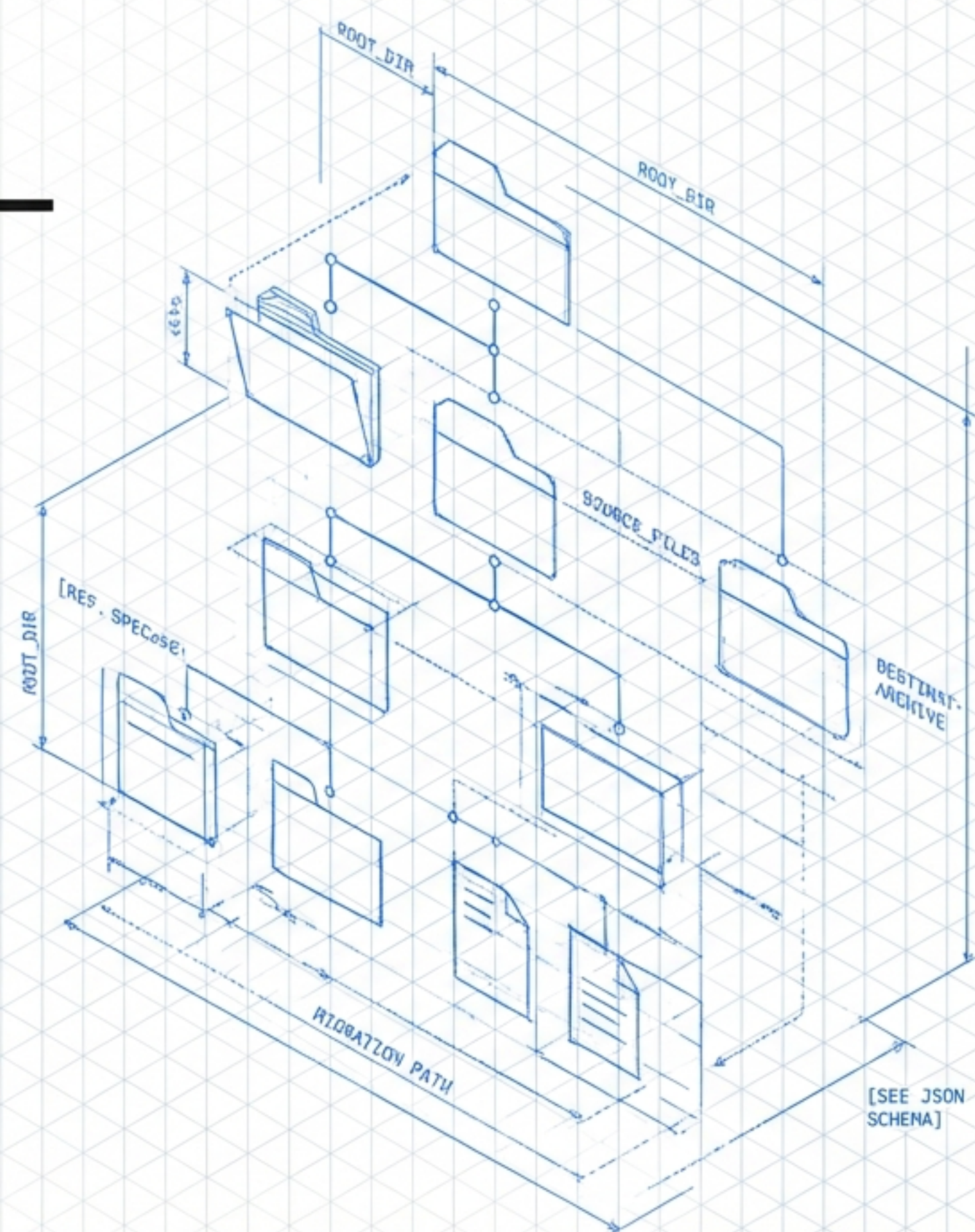


Implementation Phase 1 — Root Selection & File Migration

Evolving from Flat to
Recursive Structure

DOCUMENT VERSION	: Draft v0.1
DATE	: 2026-02-02
STATUS	: Implementation Brief (Ready)
WORKSTREAMS	: Backend (Claude Chat) + UI (Claude Code)



EXECUTIVE SUMMARY: THE SAFE EVOLUTION

This brief details the strategy to migrate from a flat file structure to a recursive issue model. The priority is to execute this evolution without breaking existing functionality or disrupting the current user experience.

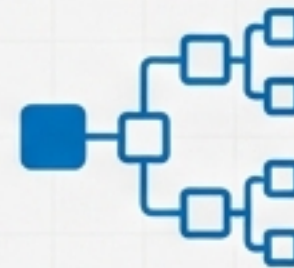
THREE PILLARS OF PHASE 1



HYBRID COMPATIBILITY

Supporting both 'node.json' (legacy) and 'issue.json' (new) simultaneously to ensure zero downtime.

CARD 1



RECURSIVE ROOTS

Moving beyond a single entry point. Any folder containing an issue file can now act as a 'root'.

CARD 2



INVISIBLE MIGRATION

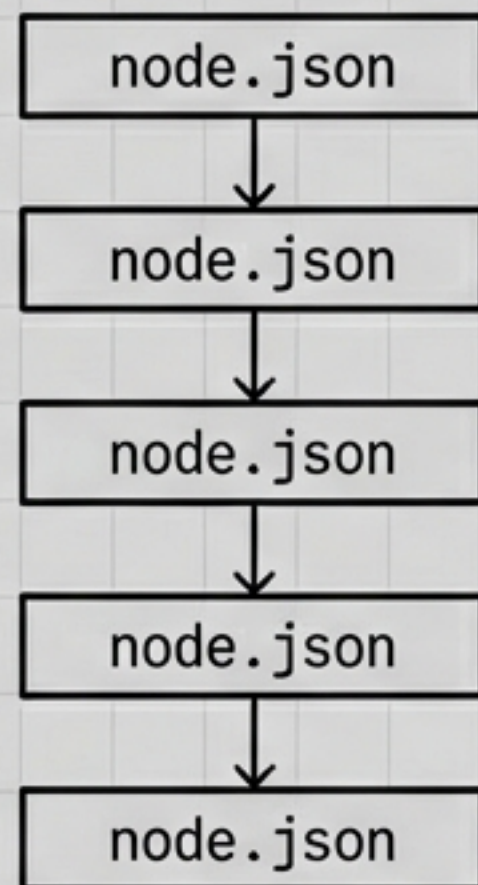
Files migrate only on 'save' events. No batch jobs. This prevents write storms and allows incremental adoption.

CARD 3

STRUCTURAL SHIFT: FROM FLAT TO RECURSIVE

Evolution without destruction of the current data model.

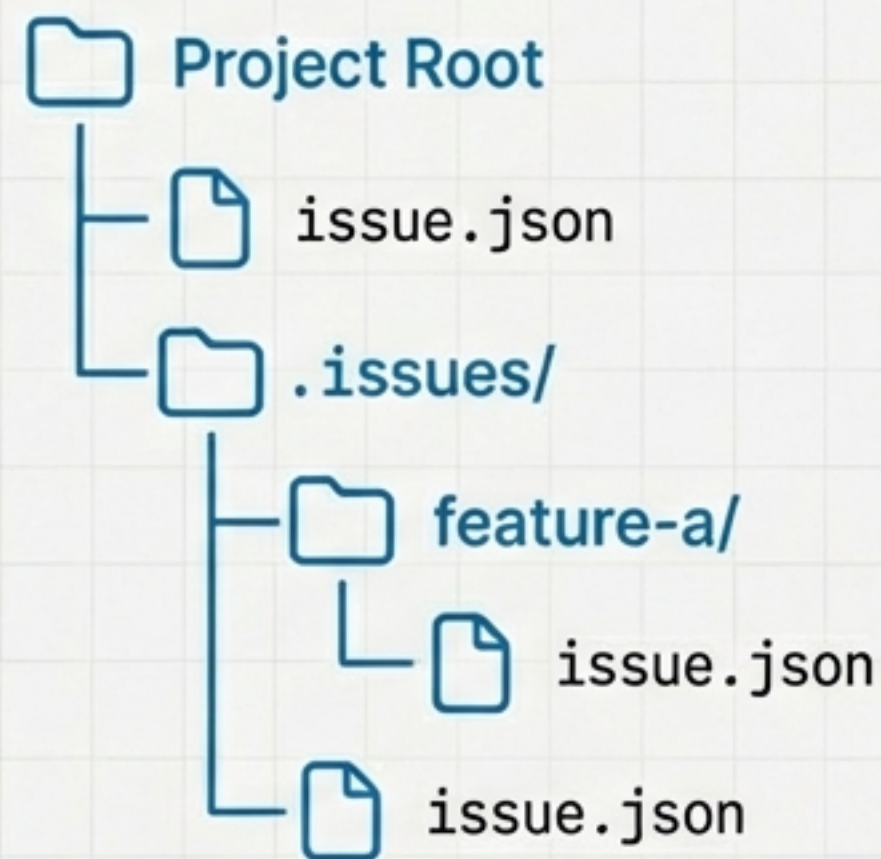
CURRENT STATE (Constrained)



Flat adjacency

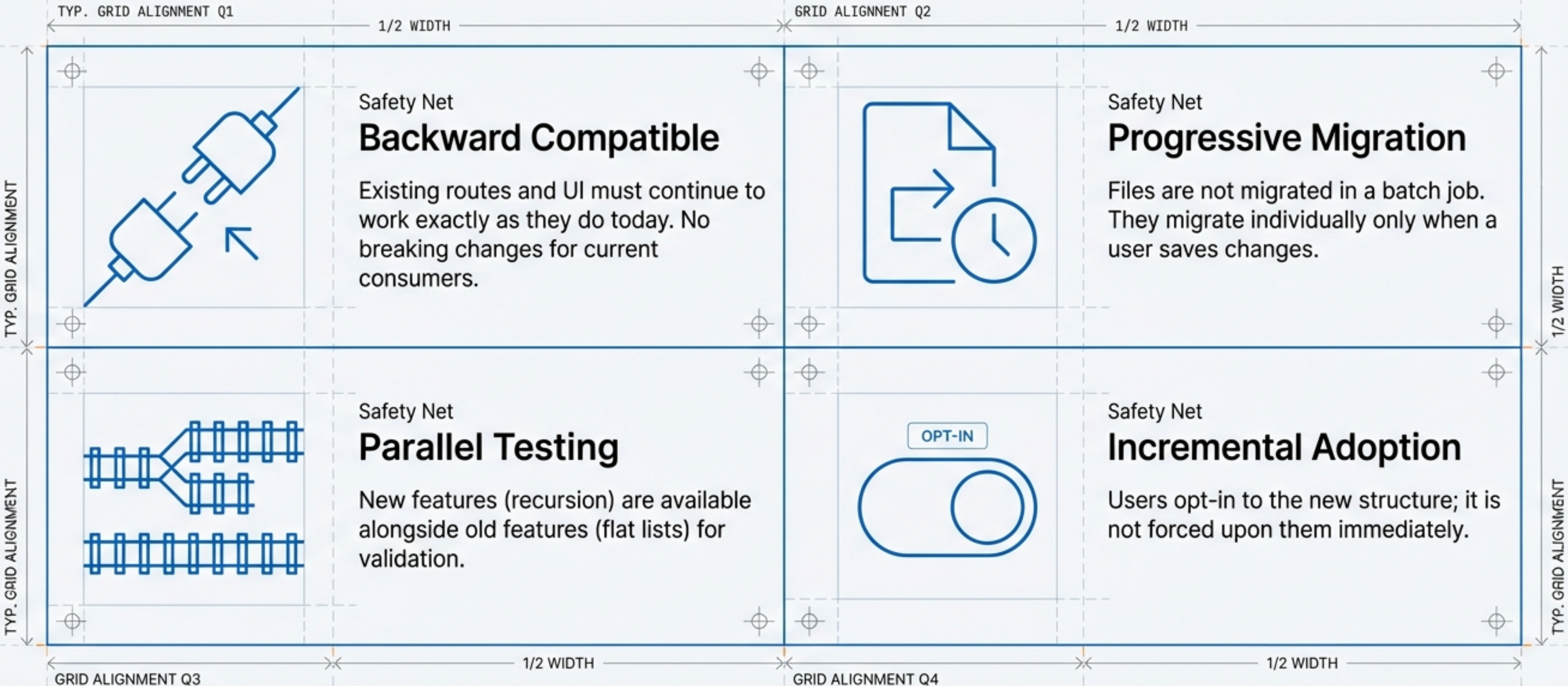
Phase 1 Transition

TARGET STATE (Flexible)



Recursive nesting

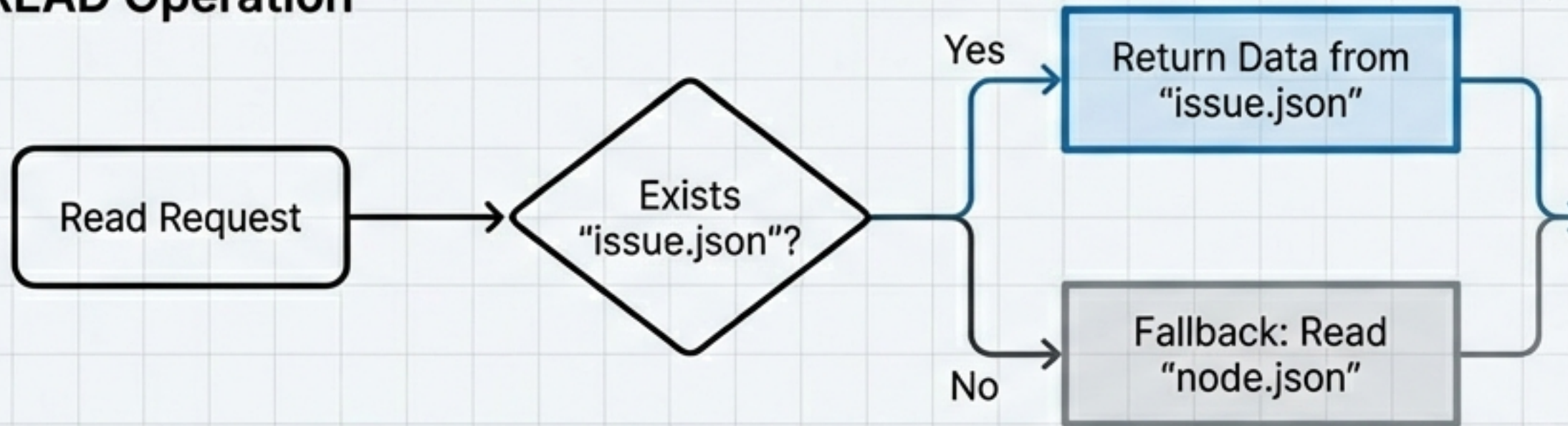
Guiding Principles for Implementation



THE HYBRID READ/WRITE LOGIC

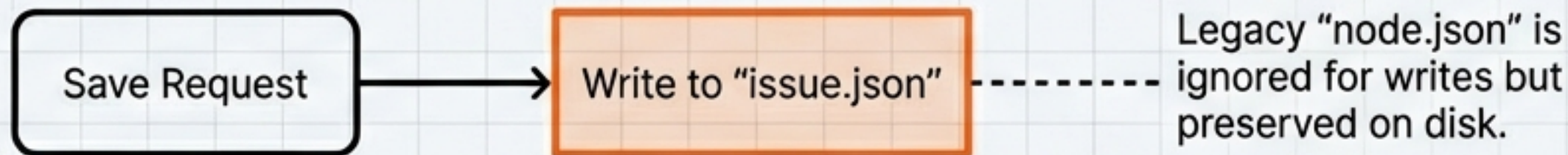
Ensuring valid data retrieval while naturally migrating formats.

READ Operation

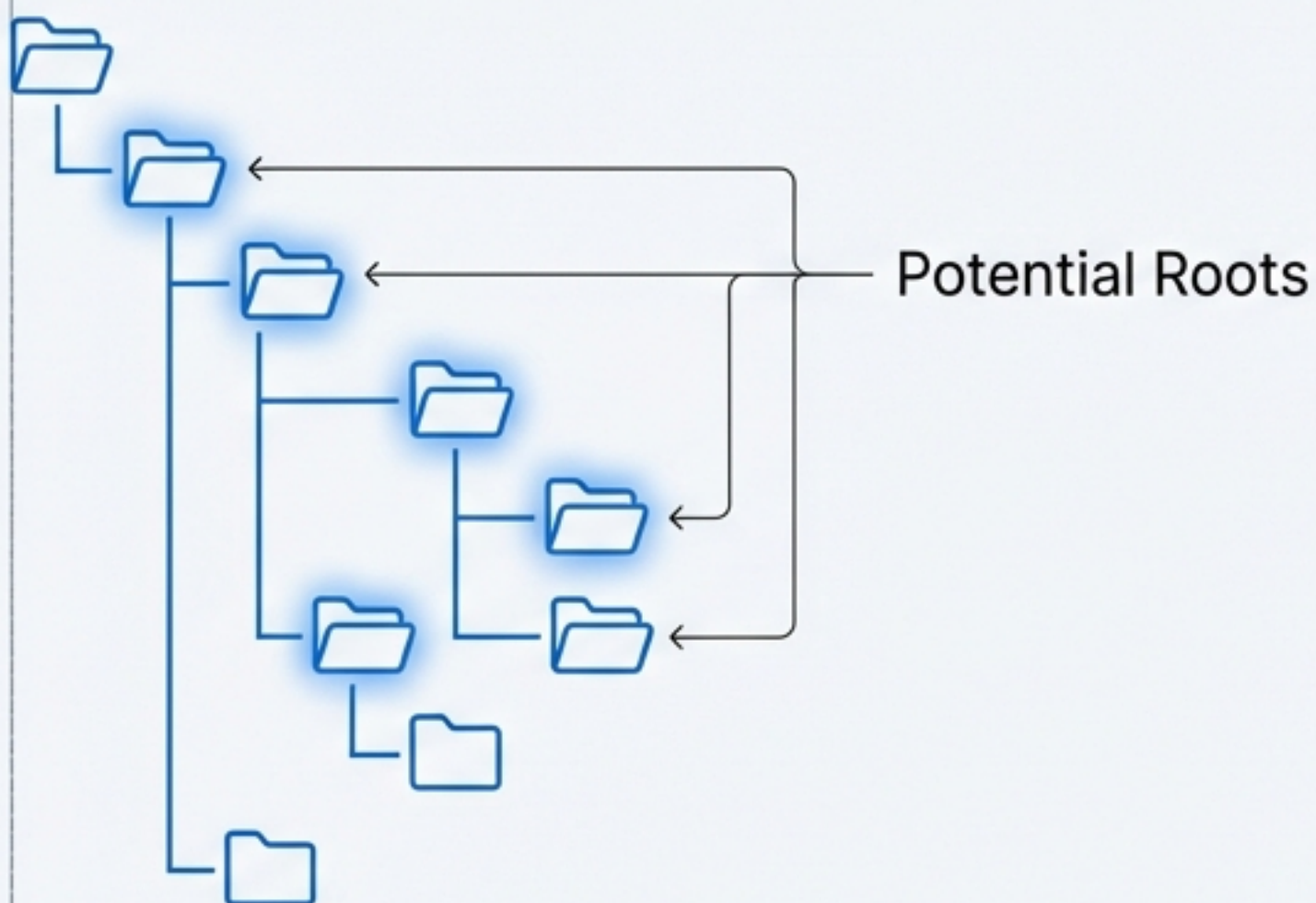


// Phase Note
Phase 1: node.json preserved
Phase 2: node.json deleted

WRITE Operation



Redefining 'Roots'



A **Root** is any folder that contains "issue.json" OR "node.json" and is selected as an entry point.

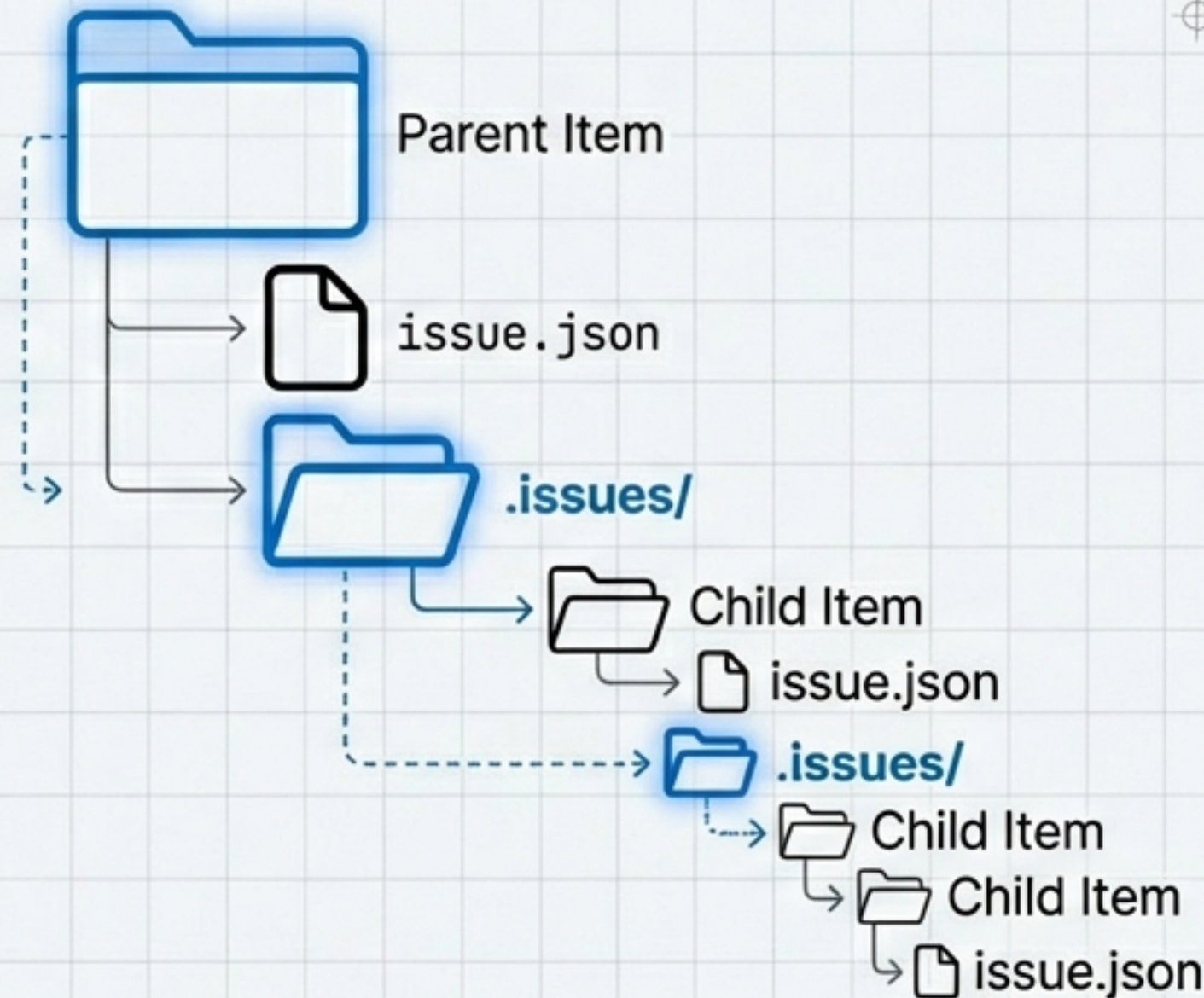
New API Endpoints

Method	Endpoint	Description
GET	/api/roots	List all available root candidates
GET	/api/roots/with-children	List roots that have issues/folders
GET	/api/roots/current	Get currently selected root
POST	/api/roots/select	Select a new root

Recursive Architecture & The 'issues/' Folder

Ensuring valid data retrieval while naturally migrating formats.

Exploded View



Creation Triggers

- User adds a child issue to a parent.
- User explicitly "converts" an issue to new structure.

The 'git-repo' Root

Introduction of a new issue type: '**git-repo**'. This acts as the uber-parent for all issues in a repository, stored in '.issues/issue.json'.

API Strategy: Stability & Extension

EXTENSION (New Capabilities)

- **POST** /api/issues/{path}/children
- **POST** /api/issues/{path}/convert
- **GET** /api/issues/{path}/children

MODULAR
EXTENSION

STABILITY (Preserved)

Existing routes preserve contracts. All
'/nodes/api/*' endpoints remain unchanged.

STABLE
FOUNDATION

Internal logic switches to read
'issue.json' OR 'node.json'.

PROJECT:
API EVOLUTION STRATEGY

DRAWING:
03

SCALE
NTS

SCALE
—

DATE
2024.05.20

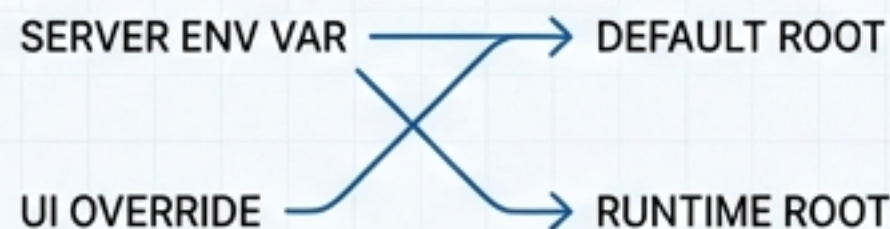
1/2 WIDTH

Key Design Decisions

1. Root Storage

Dual Strategy

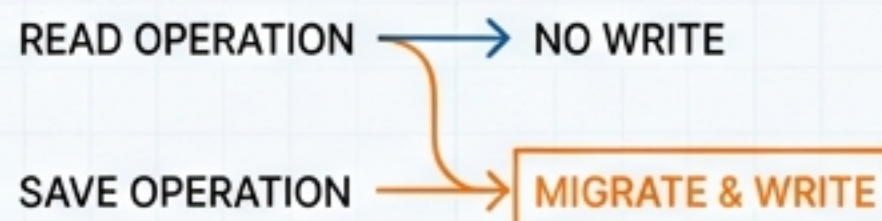
Server-wide via
'**ISSUES_ROOT_PATH**'
(Env Var) AND runtime
override via **UI Config**.



2. Auto-Migration

On Save Only

Avoids constant file writes
during read operations.
Migrates naturally during
user interaction.

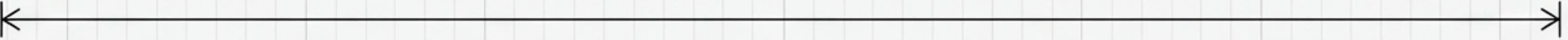


3. Root Issue Type

New Type: '**git-repo**'

Creates a natural,
semantic root for the
directory structure to
house the '.issues/' folder.





Backend Implementation Plan (Claude Chat)

Prioritized Task List B0–B9

Priority 0 (Critical)

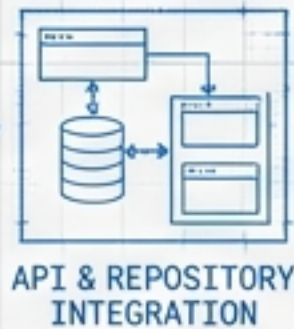
- B0: Add 'git-repo' issue type to 'node-types.json'
- B1: Modify repository to read both 'issue.json' and 'node.json'
- B2: Modify repository to write to 'issue.json' only
- B3: Add 'ISSUES__ROOT_PATH' env var support

Priority 1 (High)

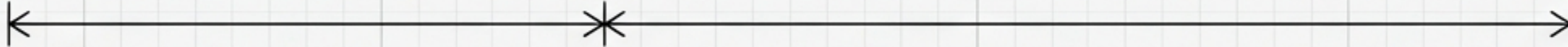
- B4: Create 'Root_Selection_Service'
- B5: Add '/api/roots' endpoints
- B6: Create root "GitRepo-1" issue in '.issues/issue.json'

Priority 2 (Standard)

- B7: Add child methods
- B8: Add children endpoints
- B9: Add conversion logic



FUTURE PHASE →



PROJECT			
RECURSIVE ISSUES ARCHITECTURE			
DRAWING			
BACKEND IMPLEMENTATION PLAN			
SCALE	DATE	DRAWING	REV
NTS	2024.05.20	04	01

UI Implementation Plan (Claude Code)

Prioritized Task List U1–U5

Priority 1 (High)

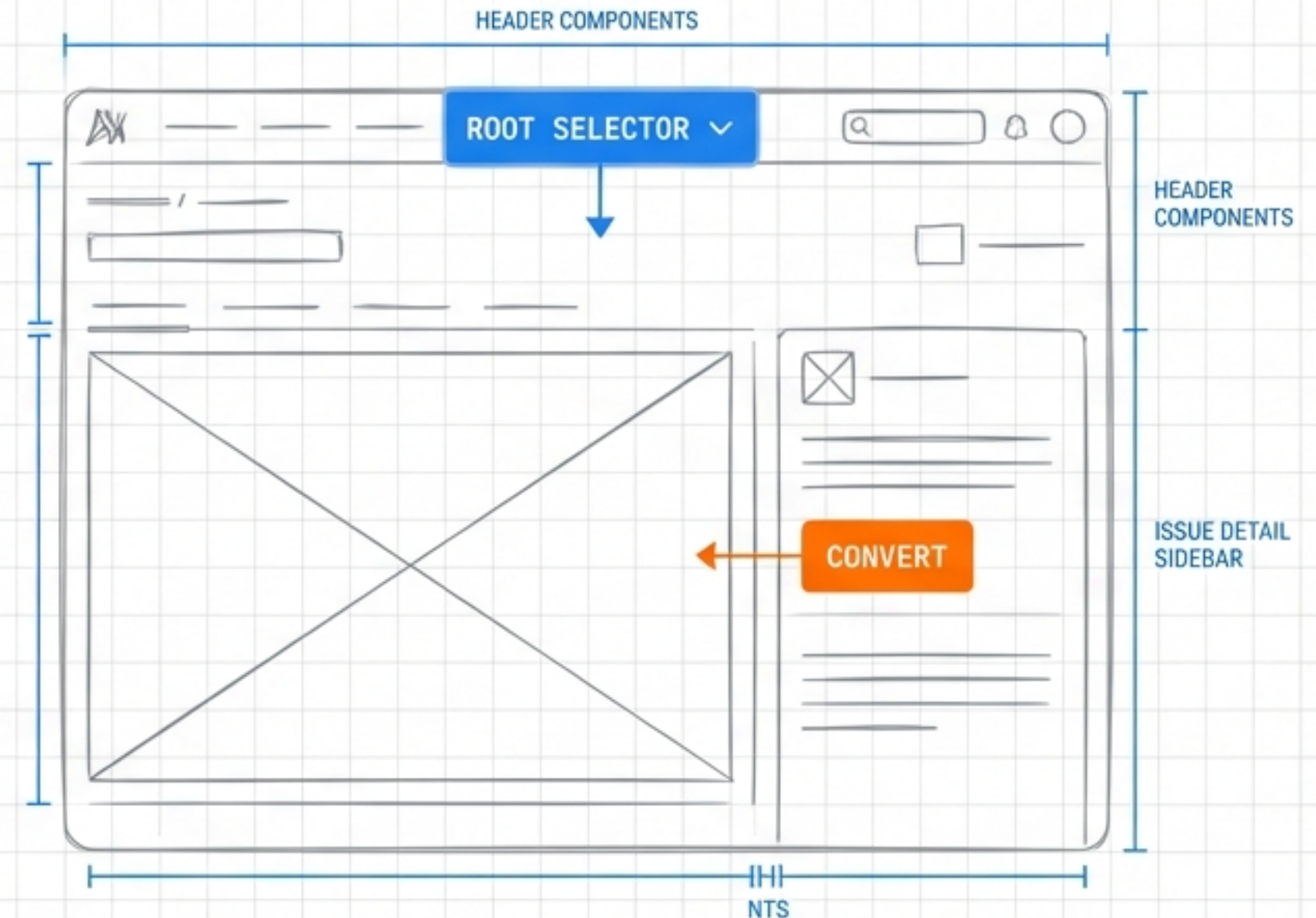
- U1: Add “Select Root” dropdown/modal in header
- U2: Display current root context indicator

Priority 2 (Standard)

- U3: Add “Convert to New Structure” button on issue detail
- U4: Add “Add Child Issue” UI for issues with “issues/” folder

Priority 3 (Low)

- U5: Tree view showing nested structure



Testing Strategy

Backend Verification

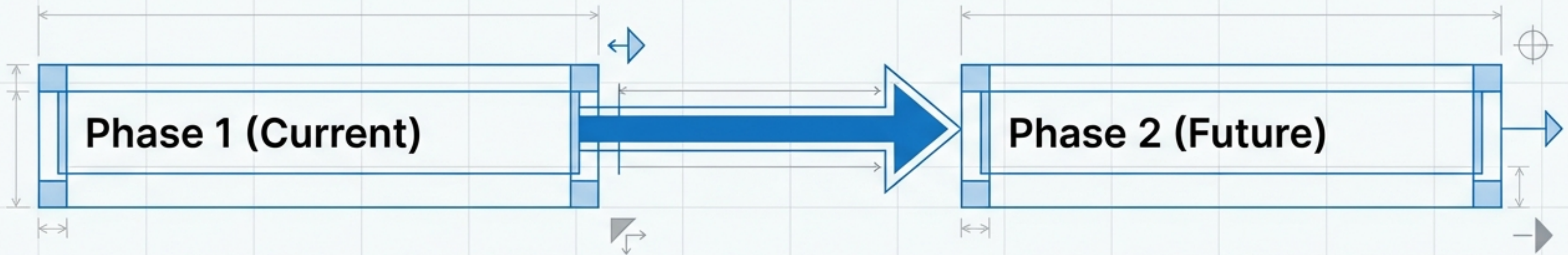
- ✓ Verify read logic handles both "node.json" and "issue.json"
- ✓ Verify write logic creates "issue.json" (and ignores old)
- ✓ Verify "/api/roots" returns valid candidates

UI Verification

- ✓ Root selector correctly populates and switches contexts
- ✓ "Convert" button successfully triggers '.issues/' folder creation
- ✓ Child issues appear correctly in the tree view hierarchy

PROJECT: RECURSIVE ISSUES ARCHITECTURE			
DRAWING: TESTING STRATEGY			
SCALE: NTS	DATE: 2024.05.22	DRAWING: 06	REV: 01

File Structure Roadmap



- **'node.json'**: Read supported, not written
- **'issue.json'**: Introduced
- **'node-types.json'**: In use

- ~~'node.json'~~: Deprecated and deleted
- **'node-types.json'**: Renamed to **'issue-types.json'**
- Client specifies root per-request



PROJECT RECURSIVE ISSUES ARCHITECTURE			
DRAWING FILE STRUCTURE ROADMAP			
SCALE NTS	DATE 2024.05.24	DRAWING NO 08	REV 02

Success Criteria

Definition of Done

- [] Backend reads both 'node.json' and 'issue.json'
- [] Backend writes ONLY to 'issue.json'
- [] '/api/roots' endpoint returns valid candidates
- [] UI shows root selection option
- [] At least one issue can be 'converted' to new structure
- [] Child issues can be created in 'issues/' folder
- [] **CRITICAL:** Existing functionality remains unchanged (all current tests pass)

PROJECT: RECURSIVE ISSUES ARCHITECTURE			
DRAWING: SUCCESS CRITERIA			
SCALE: NTS	DATE: 2024.05.28	DRAWING NO: 09	REV: 03

Ready for Implementation



STATUS: GREEN LIGHT

The plan minimizes risk through hybrid compatibility while enabling the necessary architectural evolution.

NEXT ACTION: Handover to Workstreams

- Backend: Claude Chat (Tasks B0-B9)
- UI: Claude Code (Tasks U1-U5)



PROJECT: RECURSIVE ISSUES ARCHITECTURE			
BRASIND: IMPLEMENTATION PLAN			
SCALE: NTS	DATE: 2024.05.29	DRAWING NO: 10	REV: 01